

Yujia Wang

Pronouns: She/Her/Hers

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Research Vision and Interests

My ultimate goal is to build the **next generation of collaborative intelligence systems**. Just as the mobile internet reshaped how information is shared, my research leverages advanced optimization theory to construct foundational infrastructure for collaborative learning. This will guide massive AI edge devices to develop collaboratively in an efficient and trustworthy manner, ultimately fostering a unified, principled, and intelligent future aligned with human values.

My primary research interests include theories and applications of **collaborative** and **privacy-preserving machine learning** and **optimization**. Currently, my research focuses on:

- Efficient and scalable federated learning in heterogeneous networks
- Generative methodologies in federated and collaborative learning
- Optimization methods tailored for the era of generative AI

Education

The Pennsylvania State University Ph.D. in Informatics, College of Information Sciences and Technology <i>Advisor: Dr. Jinghui Chen</i>	PA, USA 2022.01-Present
University of Michigan M.S. in Applied Statistics, Department of Statistics	MI, USA 2019.08-2021.05
Sichuan University B.S. in Statistics, College of Mathematics	Chengdu, China 2015.09-2019.06

Publications

(* indicates equal contribution)

Under Reviews

1. **Yujia Wang**, Fenglong Ma, and Jinghui Chen. Stragglers Can Contribute More: Uncertainty-Aware Distillation for Asynchronous Federated Learning. *Submitted*.
2. Tiejun Chen*, Yuanpu Cao*, **Yujia Wang***, Cho-Jui Hsieh, and Jinghui Chen. Federated Learning with Projected Trajectory Regularization. *Submitted*.

Peer-reviewed Conference Publications

3. Xin Yu, **Yujia Wang**, Jinghui Chen and Lingzhou Xue. AltLoRA: Towards Better Gradient Approximation in Low-Rank Adaptation with Alternating Projections. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems*. (**NeurIPS 2025**)
4. Yurui Chang*, Bochuan Cao*, **Yujia Wang**, Jinghui Chen and Lu Lin. JoPA: Explaining Large Language Model's Generation via Joint Prompt Attribution. In *the 63rd Annual Meeting of the Association for Computational Linguistics, 2025*. (**ACL Main 2025**)

5. **Yujia Wang**, Shiqiang Wang, Songtao Lu and Jinghui Chen. FADAS: Towards Federated Adaptive Asynchronous Optimization. In *Proceedings of the 41st International Conference on Machine Learning*, Vienna, Austria, 2024. **(ICML 2024)**
6. **Yujia Wang**, Yuanpu Cao, Jingcheng Wu, Ruoyu Chen, and Jinghui Chen. Tackling the Data Heterogeneity in Asynchronous Federated Learning with Cached Update Calibration. In *Proceedings of the 12th International Conference on Learning Representations*, Vienna, Austria, 2024. **(ICLR 2024)**
7. **Yujia Wang**, Lu Lin, and Jinghui Chen. Communication-Efficient Adaptive Federated Learning. In *Proceedings of the 39th International Conference on Machine Learning*, Baltimore, Maryland, USA, 2022. **(ICML 2022, 150+ citations)**
8. **Yujia Wang**, Lu Lin, and Jinghui Chen. Communication-Compressed Adaptive Gradient Method for Distributed Nonconvex Optimization. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, Virtual, 2022. **(AISTATS 2022)**
(A short version of this paper also appears on *International Workshop on Trustable, Verifiable and Auditable Federated Learning in Conjunction with AAAI 2022 (FL-AAAI-22)*.)

Journal Publications

9. **Yujia Wang**, Yuanpu Cao, and Jinghui Chen. ParaBlock: Communication-Computation Parallel Block Coordinate Federated Learning for Large Language Models. [Accepted with minor revision](#) in *Transactions on Machine Learning Research (TMLR)*, 2026.
10. **Yujia Wang**, and Jinghui Chen. On the Data Heterogeneity in Adaptive Federated Learning. In *Transactions on Machine Learning Research (TMLR)*, 2024.

Workshop Papers & Doctoral Forum

11. **Yujia Wang**. Towards Efficient Federated Learning in Heterogeneous Networks. In *SIAM International Conference on Data Mining*, 2024. **(SDM Doctoral Forum, 2024)**
12. **Yujia Wang**, Pei Fang, and Jinghui Chen. Accelerating Adaptive Federated Optimization with Local Gossip Communications. In *International Workshop on Federated Learning: Recent Advances and New Challenges*, NeurIPS 2022.

Research Experiences

IBM Research – Almaden Research Intern <i>Mentor:</i> Dr. Swanand R. Kadhe	CA, USA 2025.05-2025.08
The Pennsylvania State University Research Assistant <i>Advisor:</i> Dr. Jinghui Chen	PA, USA 2022.01-Present
The Pennsylvania State University Research Intern <i>Advisor:</i> Dr. Jinghui Chen	PA, USA 2021.06-2021.12
University of Michigan Research Assistant <i>Advisor:</i> Dr. Bing Zhou	MI, USA 2020.07-2020.12
Sichuan University Research Intern	Chengdu, China 2018.03-2018.12

Advisor: Dr. Zechun Hu

Awards & Honors

SIAM International Conference on Data Mining (SDM) Travel Award	2024
Pennsylvania State University College of IST Graduate Student Travel Award	2022
International Conference on Machine Learning (ICML) Travel Grant	2022
Outstanding Project of Students' Platform for Innovation and Entrepreneurship Training Program of Sichuan University	2018
National Collegiate Mathematics Competition (Second Prize of Sichuan Province,	2017
Several awards and scholarships, Sichuan University	2015-2019

Mentoring Experience

Durva Bharatbhushan Dev <i>M.Sc., at Penn State University</i> Topic: Optimization for Large Language Models	Spring 2025-Present
Jingcheng Wu <i>M.Sc., at Carnegie Mellon University</i> Currently at Google Topic: Asynchronous Federated Learning	Spring 2023-Fall 2023
Ruoyu Chen <i>M.Sc., at Carnegie Mellon University</i> Currently at Observe, Inc. Topic: Asynchronous Federated Learning	Spring 2023-Fall 2023

Presentations

Minisymposium Talk, SIAM New York-New Jersey-Pennsylvania Section Conference, Penn State University	2025
Data Engineering Tech Talks, IBM Research	2025
Poster presentation at SDM Doctoral Forum	2024
Poster presentation at AI Week Poster Session, Penn State University	2024
Poster presentation at AISTATS	2022
Poster and spotlight presentation at ICML	2022
Oral presentation at FL-AAAI	2022

Services

Conference Reviewer:	
ICML	2022, 2024, 2025, 2026
ICLR	2025, 2026
NeurIPS	2024, 2025
AISTATS	2025, 2026
KDD	2022, 2024, 2025

BigData	2023
KDD PhD Consortium	2023
AAAI	2023-2026
ECCV	2024
CVPR	2025

Journal Reviewer:

TMLR; JAIR; TMC; TNSE

Conference Volunteer:

AISTATS 2022; ICML 2022

PSU IST590 colloquium guest speaker : 2024, 2025

Teaching Experiences

Teaching Assistant for SRA 221: Overview of Information Security
The Pennsylvania State University; Spring 2023, Spring 2024

Teaching Assistant for DS 310: Machine Learning for Data Analytics
The Pennsylvania State University; Fall 2023